spin flip up excitation TDA analytic gradient

王宏博

2026.6.14

1 excited state analytic gradient basis

激发态解析梯度可以写为基态能量的梯度加激发能的梯度，即

$$\frac{dE_{exc}}{d\xi} = \frac{dE_g}{d\xi} + \frac{d\omega}{d\xi} \tag{1.1}$$

基态能量的梯度在现有程序中容易计算，这里暂时不展开，以后补充这块内容。要计算激发能的梯度，首先要得到激发能，对于 USF-TDA , Casida equation 为

$$\mathbf{AX} = \omega\mathbf{X} \tag{1.2}$$

$$\omega = \mathbf{X}^T\mathbf{AX} \tag{1.3}$$

$$A^{\beta\alpha}_{ia,jb} = \delta_{ij}F^{\alpha}_{ab} - F^{\beta}_{ij}\delta_{ab} - c_x(i^\beta j^\beta|b^\alpha a^\alpha) + f^{xc,\sigma\tau}_{pq,rs}[\rho] \tag{1.4}$$

对于 collinear kernel

$$f^{xc,\sigma\tau}_{pq,rs}[\rho] = 0 \tag{1.5}$$

对于 multicollinear kernel

$$f^{xc,\sigma\tau}_{pq,sr}[\rho] = \frac{\delta^2 E_{xc}}{\delta\rho^{\sigma}_{pq}\delta\rho^{\tau}_{sr}} \tag{1.6}$$

对于 ROKS 参考态，Casida equation 与 USF-TDA 相同，$\mathbf{A}$ 同样与 USF-TDA 相同，注意 Fock 与 USF-TDA 不同，后面 Lagrange Multiplier method 中的约束也与 USF-TDA 不同。

$$\mathbf{AX} = \omega\mathbf{X} \tag{1.7}$$

$$\omega = \mathbf{X}^T\mathbf{AX} \tag{1.8}$$

$$A^{\beta\alpha}_{ia,jb} = \delta_{ij}F^{\alpha}_{ab} - F^{\beta}_{ij}\delta_{ab} - c_x(i^\beta j^\beta|b^\alpha a^\alpha) + f^{xc,\sigma\tau}_{pq,rs}[\rho] \tag{1.9}$$

同样对于 collinear kernel

$$f^{xc,\sigma\tau}_{pq,rs}[\rho] = 0 \tag{1.10}$$

对于 multicollinear kernel

$$f^{xc,\sigma\tau}_{pq,sr}[\rho] = \frac{\delta^2 E_{xc}}{\delta\rho^{\sigma}_{pq}\delta\rho^{\tau}_{sr}} \tag{1.11}$$

2 UKS with USF-TDA analytic gradient

这一节推导 UKS 参考态下 spin flip up TDA 激发能的解析梯度。所以后面的推导中所有 $i,j \in C$ ，$a,b \in V$。与 spin conversing U-TDDFT 的解析梯度类似，同样使用 Lagrangian Multiplier method ，同样的三个约束需要满足

1. 正交（由于考虑 TDA ，所以仅剩 $\mathbf{X}$）

$$\mathbf{X}^T\mathbf{X} = 1 \tag{2.1}$$

2. Brillouin 定理

$$F^\sigma_{ia} = F^\sigma_{ai} = 0 \tag{2.2}$$

3. 正则轨道

$$S^\sigma_{pq} = \delta_{pq} \tag{2.3}$$

所以拉格朗日量可以写为

$$\mathcal{L}[\mathbf{X},\mathbf{C},\Omega,\mathbf{Z},\mathbf{W}] = \omega - \Omega(\mathbf{X}^T\mathbf{X}-1) + \sum_{ai\sigma} Z^\sigma_{ai}F^\sigma_{ai} - \sum_{pq\sigma} W^\sigma_{pq}(S^\sigma_{pq}-\delta_{pq}) \tag{2.4}$$

同样拉格朗日乘子法要求

$$\frac{\partial \mathcal{L}}{\partial X^\sigma_{ia}} = 0, \quad \frac{\partial \mathcal{L}}{\partial C^\sigma_{\mu p}} = 0, \quad \frac{\partial \mathcal{L}}{\partial \Omega} = 0, \quad \frac{\partial \mathcal{L}}{\partial Z^\sigma_{ai}} = 0, \quad \frac{\partial \mathcal{L}}{\partial W^\sigma_{pq}} = 0 \tag{2.5}$$

所以

$$\frac{d\omega}{d\xi} = \frac{\partial \mathcal{L}}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial X^\sigma_{ia}}\frac{\partial X^\sigma_{ia}}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial C^\sigma_{\mu p}}\frac{\partial C^\sigma_{\mu p}}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial \Omega}\frac{\partial \Omega}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial Z^\sigma_{ai}}\frac{\partial Z^\sigma_{ai}}{\partial \xi} + \frac{\partial \mathcal{L}}{\partial W^\sigma_{pq}}\frac{\partial W^\sigma_{pq}}{\partial \xi} = \frac{d\mathcal{L}}{d\xi} \tag{2.6}$$

同样得到拉格朗日量对坐标的偏导数，由于仅考虑 TDA 所以其中一些项无需写为对称的形式。

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \xi} = \sum_{\mu\nu}\Bigg[&\sum_\sigma (T^\sigma_{\mu\nu} + Z^{S,\sigma}_{\mu\nu})\frac{\partial h^{core}_{\mu\nu}}{\partial \xi} + \sum_\sigma (T^\sigma_{\mu\nu} + Z^{S,\sigma}_{\mu\nu})\frac{\partial g^\sigma_{\mu\nu}[\mathbf{D}]}{\partial \xi} + \sum_\sigma (T^\sigma_{\mu\nu} + Z^{S,\sigma}_{\mu\nu}) v^{xc,\sigma,(\xi)}_{\mu\nu}[\rho] \\ &- c_x \sum_{\mu\nu} X^{\beta\alpha}_{\mu\nu}\frac{\partial K^{\beta\alpha}_{\mu\nu}[\mathbf{X}]}{\partial \xi} - \sum_\sigma \frac{\partial S^\sigma_{\mu\nu}}{\partial \xi} W^\sigma_{\mu\nu}\Bigg]\end{aligned} \tag{2.7}$$

其中

$$\begin{aligned}
T^\sigma_{\mu\nu} &= \sum_{ab} C^\alpha_{\mu a} T_{ab} C^\alpha_{\nu b} + \sum_{ij} C^\beta_{\mu i} T_{ij} C^\beta_{\nu} \\
T_{ab} &= \sum_i X_{ia}X_{ib} \\
T_{ij} &= -\sum_a X_{ia}X_{ja} \\
Z^{S,\sigma}_{\mu\nu} &= \frac{1}{2}(Z^\sigma_{\mu\nu} + Z^\sigma_{\nu\mu}) \\
g^\sigma_{\mu\nu}[\mathbf{V}] &= \sum_{\kappa\lambda\tau}[(\mu\nu|\kappa\lambda) - c_x\delta_{\sigma\tau}(\mu\lambda|\kappa\nu)]V_{\kappa\lambda} \\
K^{\beta\alpha}_{\mu\nu}[\mathbf{X}] &= \sum_{\kappa\lambda}(\mu\lambda|\kappa\nu)\sum_{jb} C^\beta_{\kappa j} C^\alpha_{\lambda b} X_{jb} \\
v^{xc,\sigma,(\xi)}_{\mu\nu}[\rho] &= \frac{\partial v^{xc,\sigma}_{\mu\nu}[\rho]}{\partial \xi} \\
f^{xc,\sigma\tau,(\xi)}_{\mu\nu,\kappa\lambda}[\rho] &= \frac{\partial f^{xc,\sigma\tau}_{\mu\nu,\kappa\lambda}[\rho]}{\partial \xi}
\end{aligned} \tag{2.8}$$

同样定义

$$G^\sigma_{\mu\nu}[\mathbf{V}] = g^\sigma_{\mu\nu}[\mathbf{V}] + f^{xc,\sigma}_{\mu\nu}[\mathbf{V}] \tag{2.9}$$

同样计算 Q^{σ}_{pq}

$$
\begin{aligned}
Q^{\alpha'}_{i'j'} &= 2G_{i'j'}[\mathbf{T}] \\
Q^{\alpha'}_{a'b'} &= 2\left(\sum_a T_{aa'}F^{\alpha'}_{ab'} - c_x\sum_i X_{ia'}K_{ib'}[\mathbf{X}]\right) \\
Q^{\alpha'}_{i'a'} &= 2G_{i'a'}[\mathbf{T}] \\
Q^{\alpha'}_{a'i'} &= 2\left(\sum_a T_{aa'}F^{\alpha'}_{ai'} - c_x\sum_i X_{ia'}K_{ii'}[\mathbf{X}]\right) \\
Q^{\beta'}_{i'j'} &= 2\left(\sum_j T_{i'j}F^{\beta'}_{j'j} + G^{\beta'}_{i'j'}[\mathbf{T}] - c_x\sum_a X_{i'a}K_{j'a}[\mathbf{X}]\right) \\
Q^{\beta'}_{a'b'} &= 0 \\
Q^{\beta'}_{i'a'} &= 2\left(\sum_j T_{i'j}F^{\beta'}_{a'j} + G^{\beta'}_{i'a'}[\mathbf{T}] - c_x\sum_a X_{i'a}K_{a'a}[\mathbf{X}]\right) \\
Q^{\beta'}_{a'i'} &= 0
\end{aligned}
\tag{2.10}
$$

然后给出相同形式的 $\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C^{\sigma'}_{\mu'p'}}C^{\sigma'}_{\mu'q'}$ ，其中 σ 取 α 和 β

$$
\begin{aligned}
\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C^{\sigma'}_{\mu'i'}}C^{\sigma'}_{\mu'j'} &= Q^{\sigma'}_{i'j'} + 2G^{\sigma'}_{i'j'}[\mathbf{Z}^S] - 2W^{\sigma'}_{i'j'} \\
\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C^{\sigma}_{\mu'a'}}C^{\sigma'}_{\mu'b'} &= Q^{\sigma'}_{a'b'} - 2W^{\sigma'}_{a'b'} \\
\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C^{\sigma'}_{\mu'i'}}C^{\sigma'}_{\mu'a'} &= Q^{\sigma'}_{i'a'} + \sum_a Z^{\sigma'}_{ai'}F^{\sigma'}_{a'a} + 2G^{\sigma'}_{i'a'}[\mathbf{Z}^S] - 2W^{\sigma'}_{i'a'} \\
\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C^{\sigma'}_{\mu'a'}}C^{\sigma'}_{\mu'i'} &= Q^{\sigma'}_{a'i'} + \sum_i Z^{\sigma'}_{a'i}F^{\sigma'}_{i'i} - 2W^{\sigma'}_{a'i'}
\end{aligned}
\tag{2.11}
$$

相同形式的 Z vector equation

$$
-(Q^{\sigma'}_{i'a'} - Q^{\sigma'}_{a'i'}) = \sum_a Z^{\sigma'}_{ai'}F^{\sigma'}_{a'a} + 2G^{\sigma'}_{i'a'}[\mathbf{Z}^S] - \sum_i Z^{\sigma'}_{a'i}F^{\sigma'}_{i'i}
\tag{2.12}
$$

相同形式的 W^{σ}_{pq}

$$
\begin{aligned}
W^{\sigma'}_{i'j'} &= \frac{1}{2}Q^{\sigma'}_{i'j'} + G^{\sigma'}_{i'j'} \\
W^{\sigma'}_{a'b'} &= \frac{1}{2}Q^{\sigma'}_{a'b'} \\
W^{\sigma'}_{i'a'} &= \frac{1}{2}Q^{\sigma'}_{i'a'} + \frac{1}{2}\sum_a Z^{\sigma'}_{ai'}F^{\sigma'}_{a'a} + G^{\sigma'}_{i'a'} \\
W^{\sigma'}_{a'i'} &= \frac{1}{2}Q^{\sigma'}_{a'i'} + \frac{1}{2}\sum_i Z^{\sigma'}_{a'i}F^{\sigma'}_{i'i}
\end{aligned}
\tag{2.13}
$$

3 ROKS with USF-TDA analytic gradient

结合前一节和 spin conversing excitation 中 ROKS 参考态下的推导，这一节的公式基本都能顺利得到，使用 Lagrangian Multiplier method ，同样有三个约束

1. 正交（由于考虑 TDA ，所以仅剩 $\mathbf{X}$ ）

$$
\mathbf{X}^T\mathbf{X} = 1
\tag{3.1}
$$

2. Brillouin 定理发生变化

$$F_{ai}^{\alpha}+F_{ai}^{\beta}=0\quad F_{at}^{\alpha}=0\quad F_{it}^{\beta}=0 \tag{3.2}$$

3. 正则轨道

$$S_{pq}=\delta_{pq} \tag{3.3}$$

所以拉格朗日量可以写为

$$\mathcal{L}[\mathbf{X},\mathbf{C},\Omega,\mathbf{Z},\mathbf{W}]=\omega-\Omega(\sum_{ia}X_{ia}X_{ia}-1)+\sum_{ai}Z_{ai}(F_{ai}^{\alpha}+F_{ai}^{\beta})+\sum_{at}Z_{at}F_{at}^{\alpha}+\sum_{ti}Z_{ti}F_{ti}^{\beta}-\sum_{pq}W_{pq}(S_{pq}-\delta_{pq}) \tag{3.4}$$

同样拉格朗日乘子法要求

$$\frac{\partial\mathcal{L}}{\partial X_{ia}}=0,\quad \frac{\partial\mathcal{L}}{\partial C_{\mu p}}=0,\quad \frac{\partial\mathcal{L}}{\partial\Omega}=0,\quad \frac{\partial\mathcal{L}}{\partial Z_{ai}}=0,\quad \frac{\partial\mathcal{L}}{\partial W_{pq}}=0 \tag{3.5}$$

所以

$$\frac{d\omega}{d\xi}=\frac{\partial\mathcal{L}}{\partial\xi}+\frac{\partial\mathcal{L}}{\partial X_{ia}^{\sigma}}\frac{\partial X_{ia}^{\sigma}}{\partial\xi}+\frac{\partial\mathcal{L}}{\partial C_{\mu p}^{\sigma}}\frac{\partial C_{\mu p}^{\sigma}}{\partial\xi}+\frac{\partial\mathcal{L}}{\partial\Omega}\frac{\partial\Omega}{\partial\xi}+\frac{\partial\mathcal{L}}{\partial Z_{ai}^{\sigma}}\frac{\partial Z_{ai}^{\sigma}}{\partial\xi}+\frac{\partial\mathcal{L}}{\partial W_{pq}^{\sigma}}\frac{\partial W_{pq}^{\sigma}}{\partial\xi}=\frac{d\mathcal{L}}{d\xi} \tag{3.6}$$

同样得到拉格朗日量对坐标的偏导数，由于仅考虑 TDA 所以其中一些项无需写为对称的形式。

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial\xi}=\sum_{\mu\nu}\Bigg[&\sum_{\sigma}(T_{\mu\nu}^{\sigma}+Z_{\mu\nu}^{S})\frac{\partial h_{\mu\nu}^{core}}{\partial\xi}+\sum_{\sigma}(T_{\mu\nu}^{\sigma}+Z_{\mu\nu}^{S})\frac{\partial g_{\mu\nu}^{\sigma}[\mathbf{D}]}{\partial\xi}+\sum_{\sigma}(T_{\mu\nu}^{\sigma}+Z_{\mu\nu}^{S})v_{\mu\nu}^{xc,\sigma,(\xi)}[\rho]\\&-c_x\sum_{\mu\nu}X_{\mu\nu}^{\beta\alpha}\frac{\partial K_{\mu\nu}^{\beta\alpha}[\mathbf{X}]}{\partial\xi}-\sum_{\sigma}\frac{\partial S_{\mu\nu}}{\partial\xi}W_{\mu\nu}\Bigg]\end{aligned} \tag{3.7}$$

其中

$$\begin{aligned}T_{\mu\nu}^{\sigma}&=\sum_{ab}C_{\mu a}^{\alpha}T_{ab}C_{\nu b}^{\alpha}+\sum_{ij}C_{\mu i}^{\beta}T_{ij}C_{\nu}^{\beta}\\T_{ab}&=\sum_{i}X_{ia}X_{ib}\\T_{ij}&=-\sum_{a}X_{ia}X_{ja}\\Z_{\mu\nu}^{S}&=\frac{1}{2}(Z_{\mu\nu}+Z_{\nu\mu})=Z_{\nu\mu}^{S}\\Z_{\mu\nu}&=\sum_{ai}C_{\mu i}C_{\nu a}Z_{ai}+\sum_{at}C_{\mu t}C_{\nu a}Z_{ai}+\sum_{ti}C_{\mu t}C_{\nu i}Z_{ti}\\g_{\mu\nu}^{\sigma}[\mathbf{V}]&=\sum_{\kappa\lambda\tau}[(\mu\nu|\kappa\lambda)-c_x\delta_{\sigma\tau}(\mu\lambda|\kappa\nu)]V_{\kappa\lambda}\\K_{\mu\nu}^{\beta\alpha}[\mathbf{X}]&=\sum_{\kappa\lambda}(\mu\lambda|\kappa\nu)\sum_{jb}C_{\kappa j}^{\beta}C_{\lambda b}^{\alpha}X_{jb}\\v_{\mu\nu}^{xc,\sigma,(\xi)}[\rho]&=\frac{\partial v_{\mu\nu}^{xc,\sigma}[\rho]}{\partial\xi}\\f_{\mu\nu,\kappa\lambda}^{xc,\sigma\tau,(\xi)}[\rho]&=\frac{\partial f_{\mu\nu,\kappa\lambda}^{xc,\sigma\tau}[\rho]}{\partial\xi}\end{aligned} \tag{3.8}$$

同样定义

$$G_{\mu\nu}^{\sigma}[\mathbf{V}]=g_{\mu\nu}^{\sigma}[\mathbf{V}]+f_{\mu\nu}^{xc,\sigma}[\mathbf{V}] \tag{3.9}$$

同样计算 $\sum_{\mu'} \frac{\partial \mathcal{L}}{\partial C^{\sigma'}_{\mu' p'}} C^{\sigma'}_{\mu' q'} = 0$。首先计算 $Q^{\sigma'}_{pq}$

$$
\begin{aligned}
Q^{\alpha'}_{i'j'} &= 2G_{i'j'}[\mathbf{T}] \\
Q^{\alpha'}_{i't'} &= 2G_{i't'}[\mathbf{T}] \\
Q^{\alpha'}_{i'a'} &= 2G_{i'a'}[\mathbf{T}] \\
Q^{\alpha'}_{t'i'} &= 2G_{t'i'}[\mathbf{T}] \\
Q^{\alpha'}_{t'u'} &= 2G_{t'u'}[\mathbf{T}] \\
Q^{\alpha'}_{t'a'} &= 2G_{t'a'}[\mathbf{T}] \\
Q^{\alpha'}_{a'i'} &= 2\left(\sum_a T_{aa'} F^{\alpha'}_{ai'} - c_x \sum_i X_{ia'} K_{ii'}[\mathbf{X}]\right) \\
Q^{\alpha'}_{a't'} &= 2\left(\sum_a T_{aa'} F^{\alpha'}_{at'} - c_x \sum_i X_{ia'} K_{it'}[\mathbf{X}]\right) \\
Q^{\alpha'}_{a'b'} &= 2\left(\sum_a T_{aa'} F^{\alpha'}_{ab'} - c_x \sum_i X_{ia'} K_{ib'}[\mathbf{X}]\right) \\
Q^{\beta'}_{i'j'} &= 2\left(\sum_j T_{i'j} F^{\beta'}_{j'j} + G^{\beta'}_{i'j'}[\mathbf{T}] - c_x \sum_a X_{i'a} K_{j'a}[\mathbf{X}]\right) \\
Q^{\beta'}_{i't'} &= 2\left(\sum_j T_{i'j} F^{\beta'}_{t'j} + G^{\beta'}_{i't'}[\mathbf{T}] - c_x \sum_a X_{i'a} K_{t'a}[\mathbf{X}]\right) \\
Q^{\beta'}_{i'a'} &= 2\left(\sum_j T_{i'j} F^{\beta'}_{a'j} + G^{\beta'}_{i'a'}[\mathbf{T}] - c_x \sum_a X_{i'a} K_{a'a}[\mathbf{X}]\right) \\
Q^{\beta'}_{t'i'} &= 0 \\
Q^{\beta'}_{t'u'} &= 0 \\
Q^{\beta'}_{t'a'} &= 0 \\
Q^{\beta'}_{a'i'} &= 0 \\
Q^{\beta'}_{a't'} &= 0 \\
Q^{\beta'}_{a'b'} &= 0
\end{aligned}
\tag{3.10}
$$

然后得到

$$
\begin{aligned}
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu'i'}}C_{\mu'j'}=Q_{i'j'}+2G_{i'j'}[\mathbf{Z}^S]-2W_{i'j'}\\
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu'a'}}C_{\mu'b'}=Q_{a'b'}-2W_{a'b'}\\
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu't'}}C_{\mu'u'}=Q_{t'u'}+2G^{\alpha'}_{t'u'}[\mathbf{Z}^S]-2W_{t'u'}\\
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu'i'}}C_{\mu'a'}=Q_{i'a'}+\sum_{a\sigma'}Z_{ai'}F^{\sigma'}_{aa'}+\sum_{t}Z_{ti'}F^{\beta'}_{ta'}+2G_{i'a'}[\mathbf{Z}^S]-2W_{i'a'}\\
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu'i'}}C_{\mu't'}=Q_{i't'}+\sum_{a\sigma'}Z_{ai'}F^{\sigma'}_{at'}+\sum_{t}Z_{ti'}F^{\beta'}_{tt'}+2G_{i't'}[\mathbf{Z}^S]-2W_{i't'}\\
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu'a'}}C_{\mu'i'}=Q_{a'i'}+\sum_{i\sigma'}Z_{a'i}F^{\sigma'}_{i'i}+\sum_{t}Z_{a't}F^{\alpha'}_{i't}-2W_{a'i'}\\
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu'a'}}C_{\mu't'}=Q_{a't'}+\sum_{i\sigma'}Z_{a'i}F^{\sigma'}_{t'i}+\sum_{t}Z_{a't}F^{\alpha'}_{t't}-2W_{a't'}\\
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu't'}}C_{\mu'i'}=Q_{t'i'}+\sum_{a}Z_{at'}F^{\alpha'}_{ai'}+\sum_{i}Z_{t'i}F^{\beta'}_{i'i}+2G^{\alpha'}_{t'i'}[\mathbf{Z}^S]-2W_{t'i'}\\
&\sum_{\mu'}\frac{\partial\mathcal{L}}{\partial C_{\mu't'}}C_{\mu'a'}=Q_{t'a'}+\sum_{a}Z_{at'}F^{\alpha'}_{aa'}+\sum_{i}Z_{t'i}F^{\beta'}_{a'i}+2G^{\alpha'}_{t'a'}[\mathbf{Z}^S]-2W_{t'a'}
\end{aligned}
\tag{3.11}
$$

对于 ROKS 有 $W_{pq}=W_{qp}$ ，同样可以得到 Z vector equation

$$
\begin{aligned}
-\left(Q_{i'a'}-Q_{a'i'}\right)&=\sum_{i\sigma'}Z_{a'i}F^{\sigma'}_{i'i}+\sum_{t}Z_{a't}F^{\alpha'}_{i't}-\sum_{a\sigma'}Z_{ai'}F^{\sigma'}_{aa'}-\sum_{t}Z_{ti'}F^{\beta'}_{ta'}-2G_{i'a'}[\mathbf{Z}^S]\\
-\left(Q_{i't'}-Q_{t'i'}\right)&=\sum_{a}Z_{at'}F^{\alpha'}_{ai'}+\sum_{i}Z_{t'i}F^{\beta'}_{i'i}+2G^{\alpha'}_{t'i'}[\mathbf{Z}^S]-\sum_{a\sigma'}Z_{ai'}F^{\sigma'}_{at'}-\sum_{t}Z_{ti'}F^{\beta'}_{tt'}-2G_{i't'}[\mathbf{Z}^S]\\
-\left(Q_{t'a'}-Q_{a't'}\right)&=\sum_{i\sigma'}Z_{a'i}F^{\sigma'}_{t'i}+\sum_{t}Z_{a't}F^{\alpha'}_{t't}-\sum_{a}Z_{at'}F^{\alpha'}_{aa'}-\sum_{i}Z_{t'i}F^{\beta'}_{a'i}-2G^{\alpha'}_{t'a'}[\mathbf{Z}^S]
\end{aligned}
\tag{3.12}
$$

求解 Z vector equation 可以得到 Z_{ai}, Z_{at}, Z_{ti} ，进而可以得到 W_{pq}

$$
\begin{aligned}
W_{i'j'}&=\frac{1}{2}Q_{i'j'}+G_{i'j'}[\mathbf{Z}^S]\\
W_{a'b'}&=\frac{1}{2}Q_{a'b'}\\
W_{t'u'}&=\frac{1}{2}Q_{t'u'}+G^{\alpha'}_{t'u'}[\mathbf{Z}^S]\\
W_{i'a'}&=\frac{1}{2}Q_{i'a'}+\frac{1}{2}\sum_{a}Z_{ai'}F^{\sigma'}_{a'a}+\frac{1}{2}\sum_{t}Z_{ti'}F^{\beta'}_{ta'}+G_{i'a'}[\mathbf{Z}^S]\\
W_{i't'}&=\frac{1}{2}Q_{i't'}+\frac{1}{2}\sum_{a\sigma'}Z_{ai'}F^{\sigma'}_{at'}+\frac{1}{2}\sum_{t}Z_{ti'}F^{\beta'}_{tt'}+G_{i't'}[\mathbf{Z}^S]\\
W_{a'i'}&=\frac{1}{2}Q_{a'i'}+\frac{1}{2}\sum_{i}Z_{a'i}F^{\sigma'}_{i'i}+\frac{1}{2}\sum_{t}Z_{a't}F^{\alpha'}_{i't}\\
W_{a't'}&=\frac{1}{2}Q_{a't'}+\frac{1}{2}\sum_{i\sigma'}Z_{a'i}F^{\sigma'}_{t'i}+\frac{1}{2}\sum_{t}Z_{a't}F^{\alpha'}_{t't}\\
W_{t'i'}&=\frac{1}{2}Q_{t'i'}+\frac{1}{2}\sum_{a}Z_{at'}F^{\alpha'}_{ai'}+\frac{1}{2}\sum_{i}Z_{t'i}F^{\beta'}_{i'i}+G^{\alpha'}_{t'i'}[\mathbf{Z}^S]\\
W_{t'a'}&=\frac{1}{2}Q_{t'a'}+\frac{1}{2}\sum_{a}Z_{at'}F^{\alpha'}_{aa'}+\frac{1}{2}\sum_{i}Z_{t'i}F^{\beta'}_{a'i}+G^{\alpha'}_{t'a'}[\mathbf{Z}^S]
\end{aligned}
\tag{3.13}
$$

有了 W_{pq} 和 Z_{pq} 带入 (3.7) ，然后加基态能量的梯度即可完成计算。